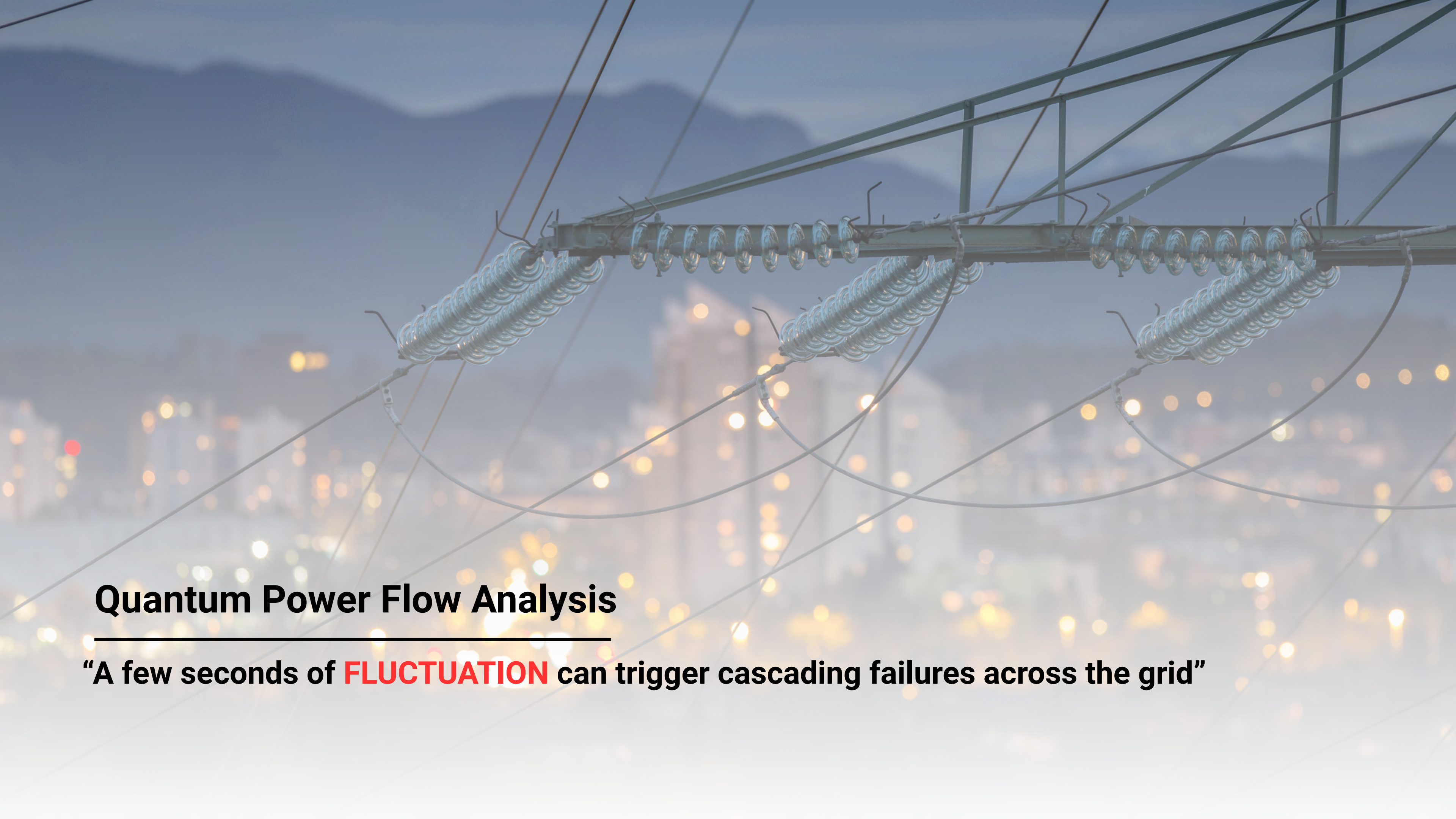


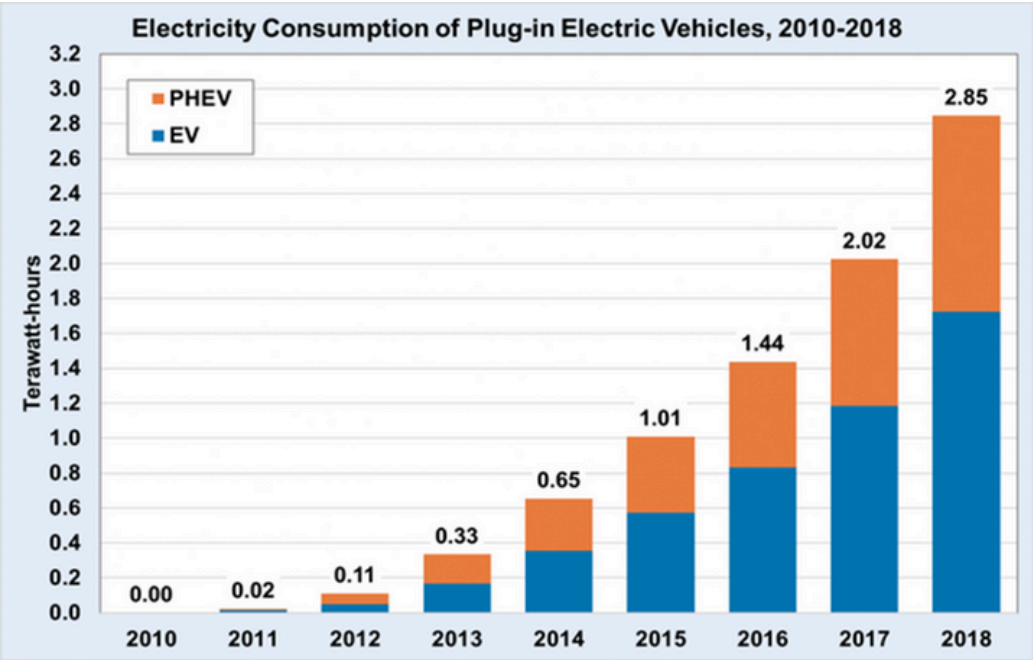


Quantum Power Flow Analysis

“A few seconds of **FLUCTUATION can trigger cascading failures across the grid”**

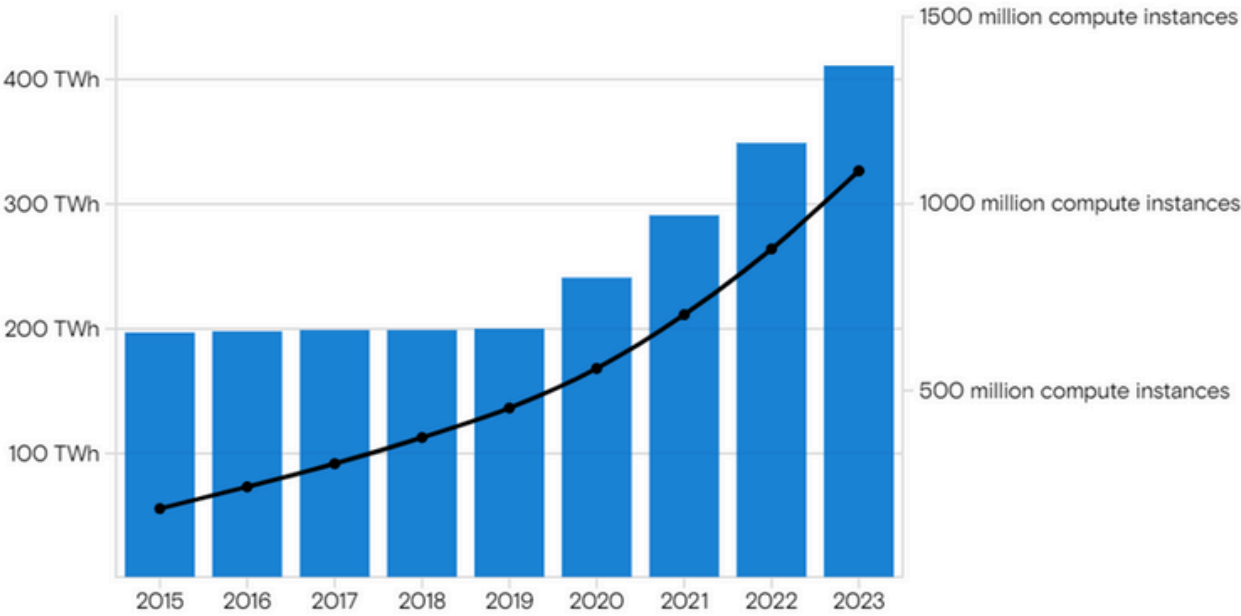
The state of energy consumption

Electric vehicles



DATA CENTER

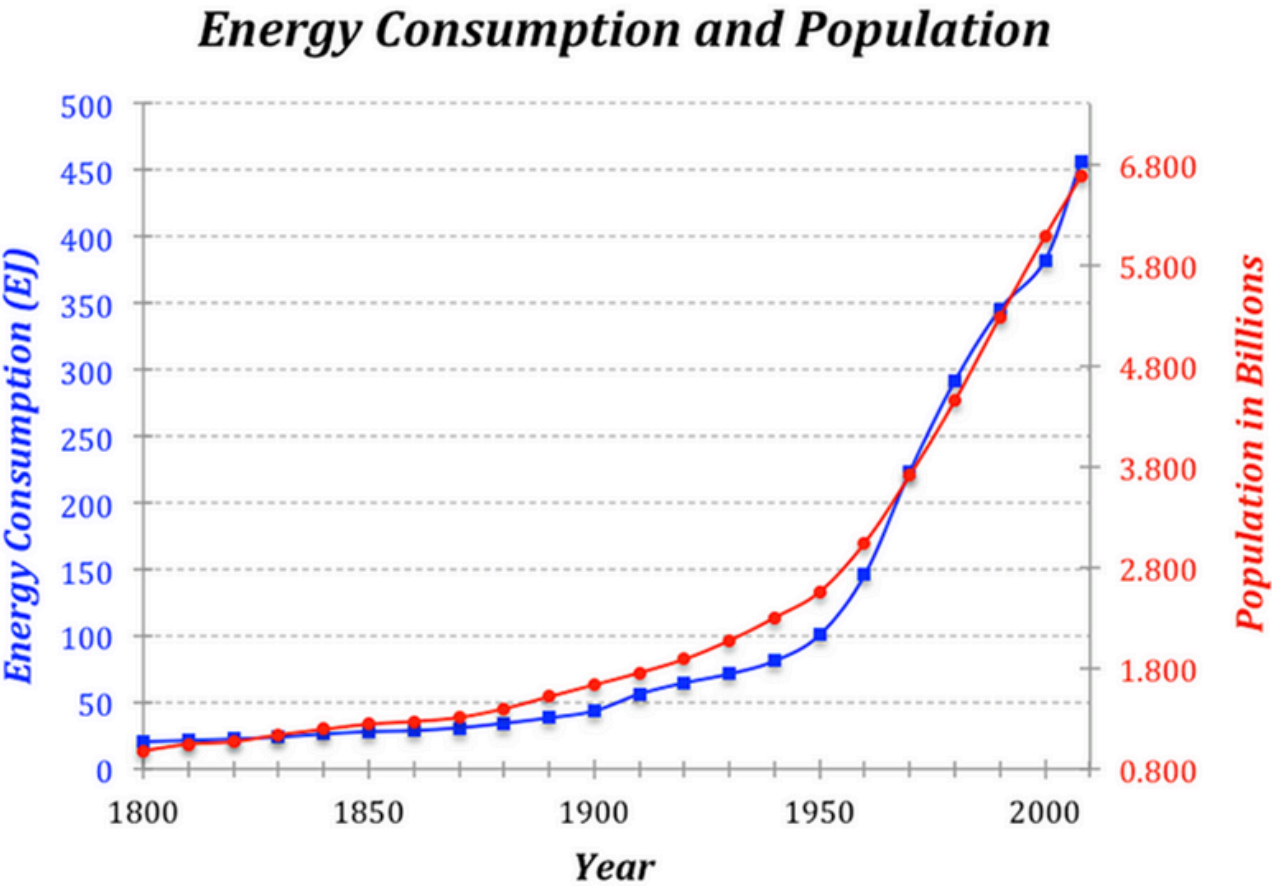
The workload demand for data centers...
...and the power they consumed



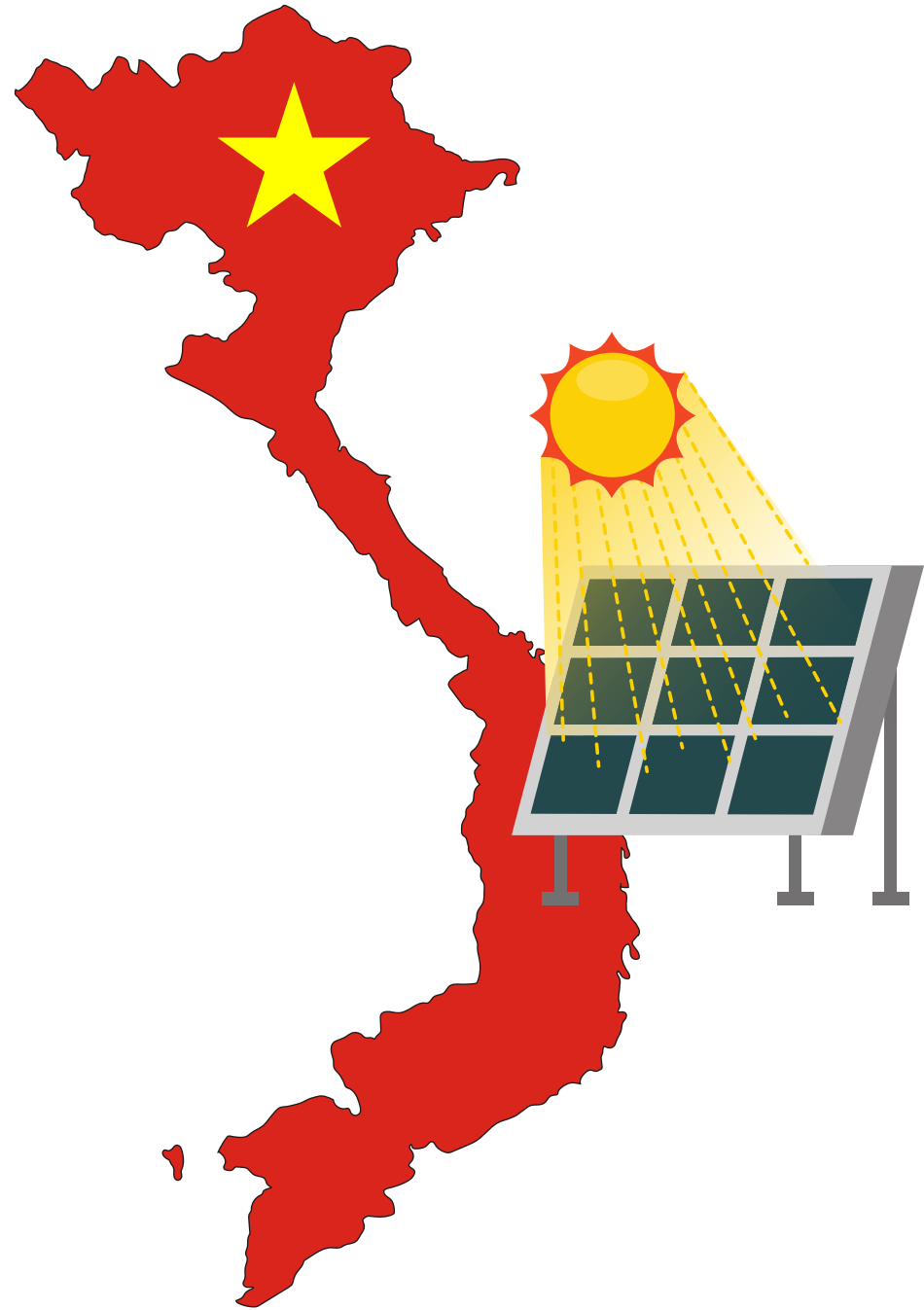
Source: Masanet et al. (2020), Cisco, IEA, Goldman Sachs Research
The data center power demand for 2023 is an estimate.

Goldman Sachs

Population



ASEAN Case Example



Vietnam

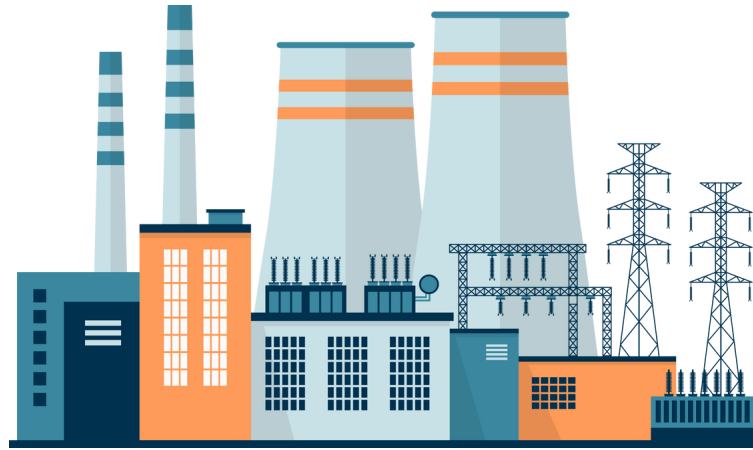
- Solar Farms (16-19 GW)
- 13.7% of Generation Mix
- Grid Stability \$33 billion

**Reducing peak-time fluctuations can
cut transmission losses by up to**

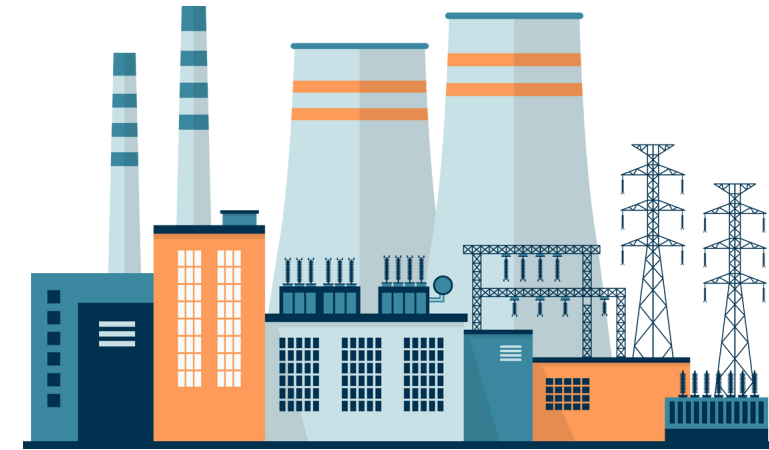
20%

boosting grid efficiency and reliability.*

Energy system

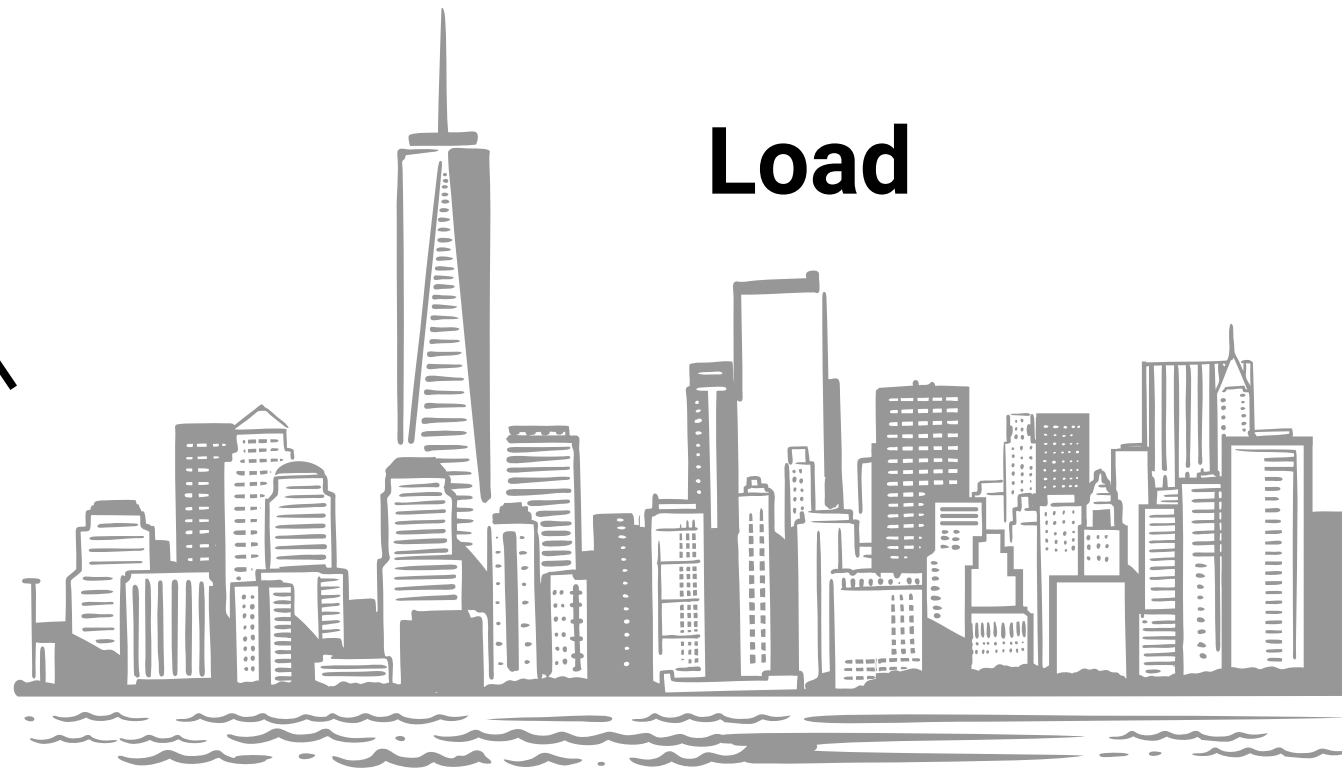


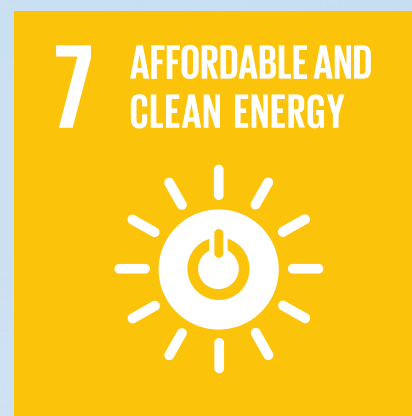
**Generator with Renewable
resources**



Generator (Swing)

Load





To meet ASEAN's 23% renewable energy goal by 2030, we need to make the energy grid more efficient through quantum-based power flow solutions

Existing Power Flow solutions vs Quantum-based

Limitations of Classical computer-based solutions

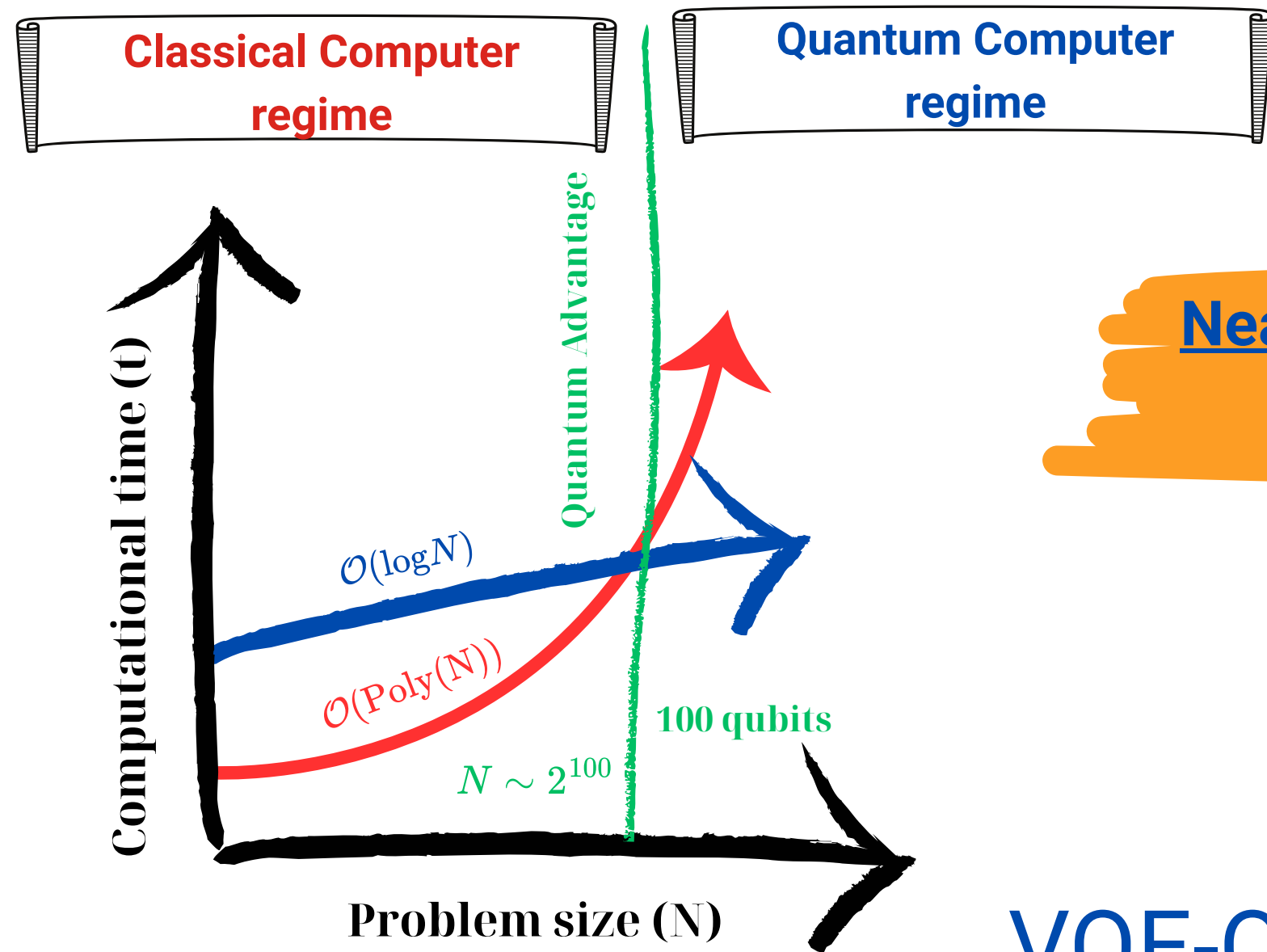
- Slow response to dynamic grid conditions
- Poor accuracy under RE variability (solar/wind)
- Limited scalability in large smart grids

Advantages of Quantum-based power flow analysis

- Real-time simulations for fast contingency planning
- Probabilistic modeling handles uncertainty effectively
- Scales efficiently with more nodes and grid complexity

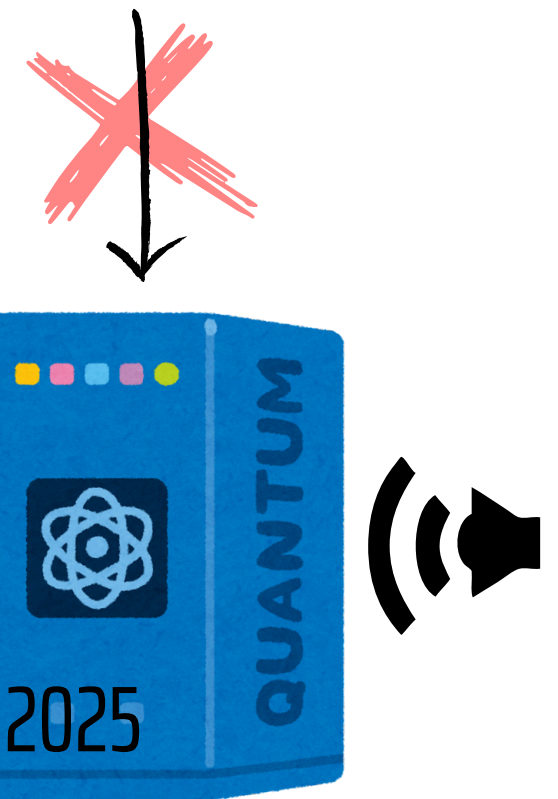
Quantum Power Flow Algorithm

Quantum Power Flow (QPF)



Near-term applications on
current devices!!

HHL algo.

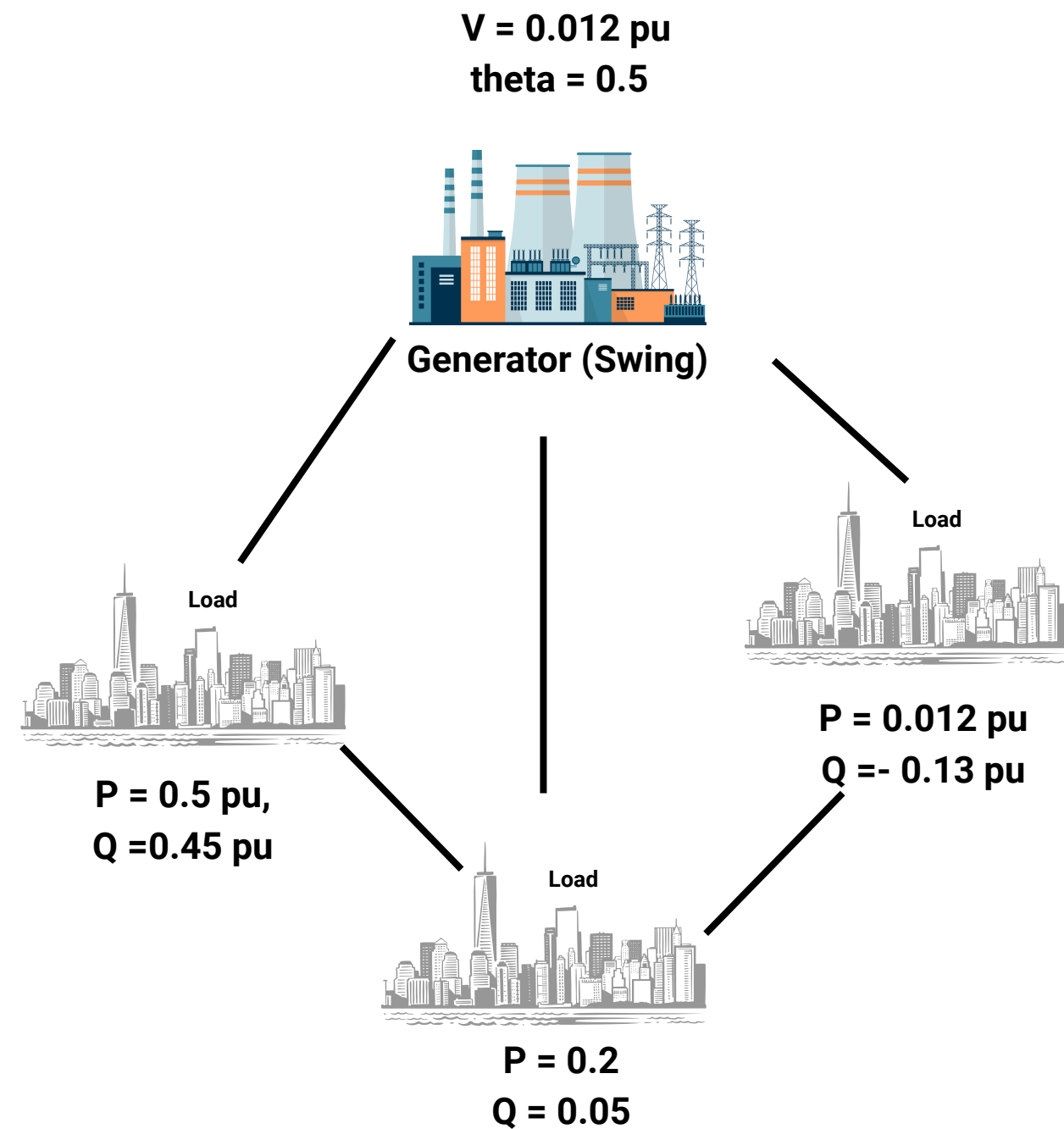


Reduce quantum
resources by half

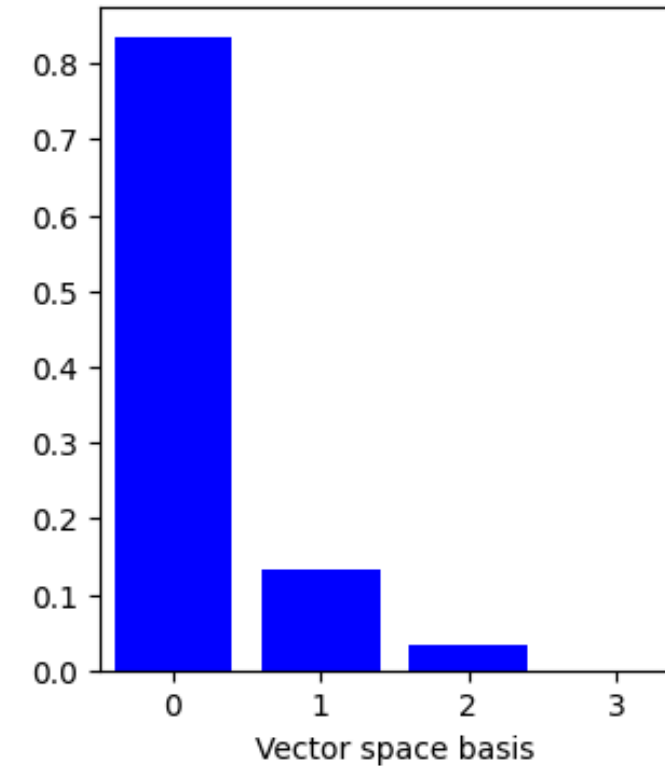
VQE-QLS algo.

VQLS algo.

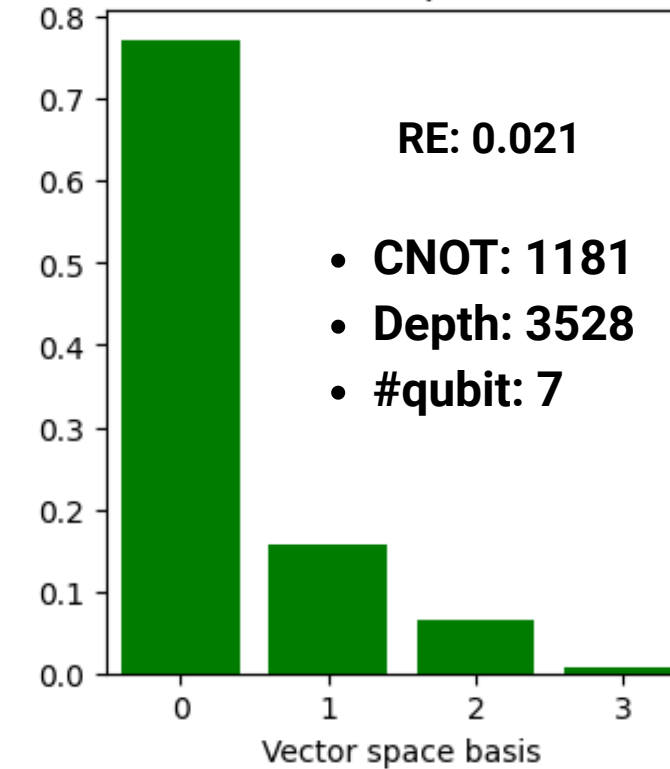
Quantum Power Flow Algorithm



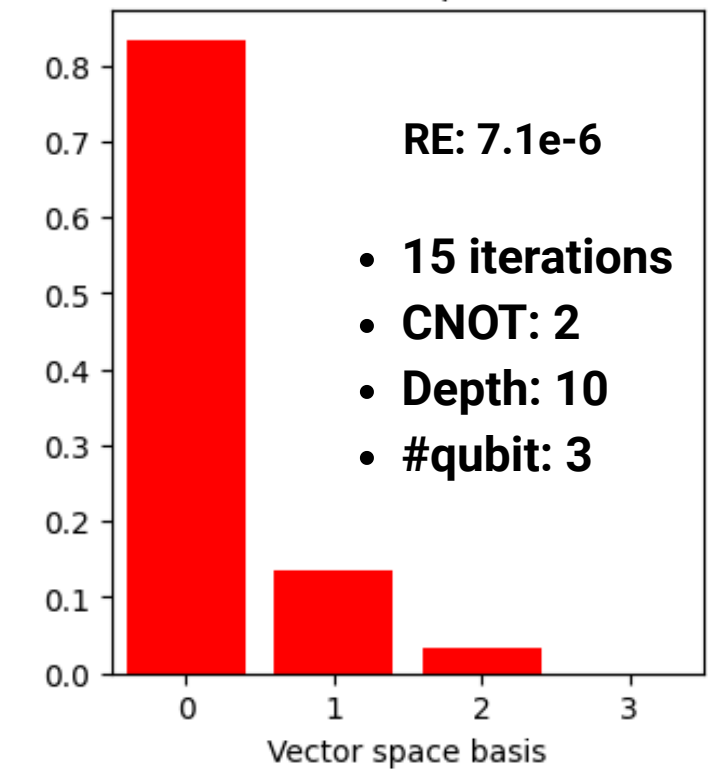
Classical probabilities



HHL Quantum probabilities



VQLS Quantum probabilities



- Comparable accuracy with the classical method
- Better than HHL (3000 times)
- Considerably reduce the quantum resources against HHL (~99%)

Market Size

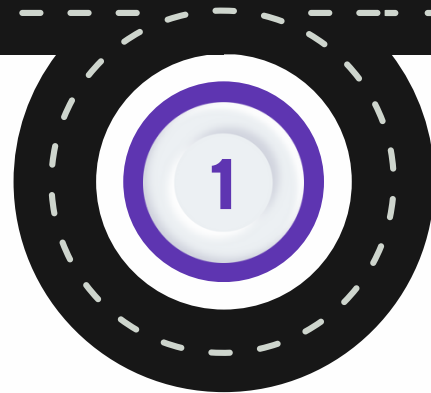
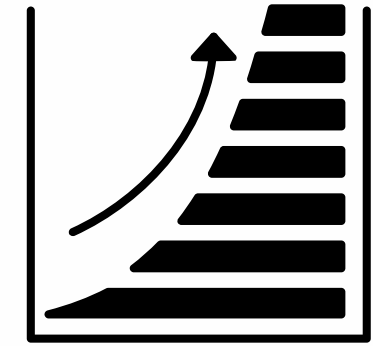
ASEAN Energy transition market
\$597.8 billion (2022)



\$1,383 billion (2032)
at **8.80%** compounded annual
growth rate (CAGR)

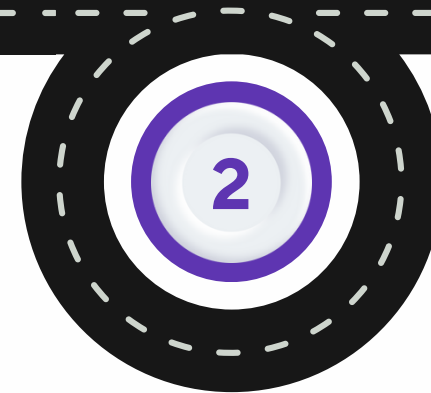


Our Roadmap



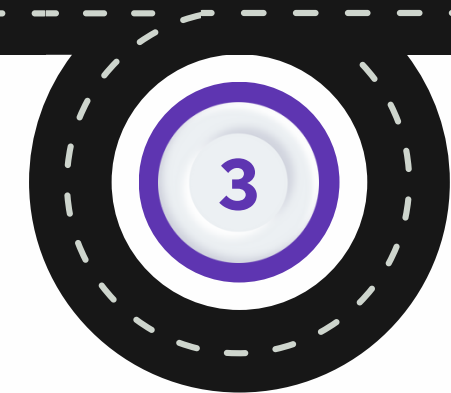
Phase 1: Foundation & Local Partnership (Years 1-2)

- Focus on a single, high-renewable-energy town in Vietnam.
- Partner with a leading Vietnamese university and the local power utility to build trust.
- Deliver a successful proof-of-concept on a real-world grid problem.
- Demonstrate a clear technological advantage.



Phase 2: Pilot Deployment & Value Demonstration (Years 3-4)

- Deploy user-friendly software in within the local utility's control center.
- Allow operators to see benefits firsthand without risk.
- Create a powerful business case that quantifies the value delivered in improved stability and financial savings.



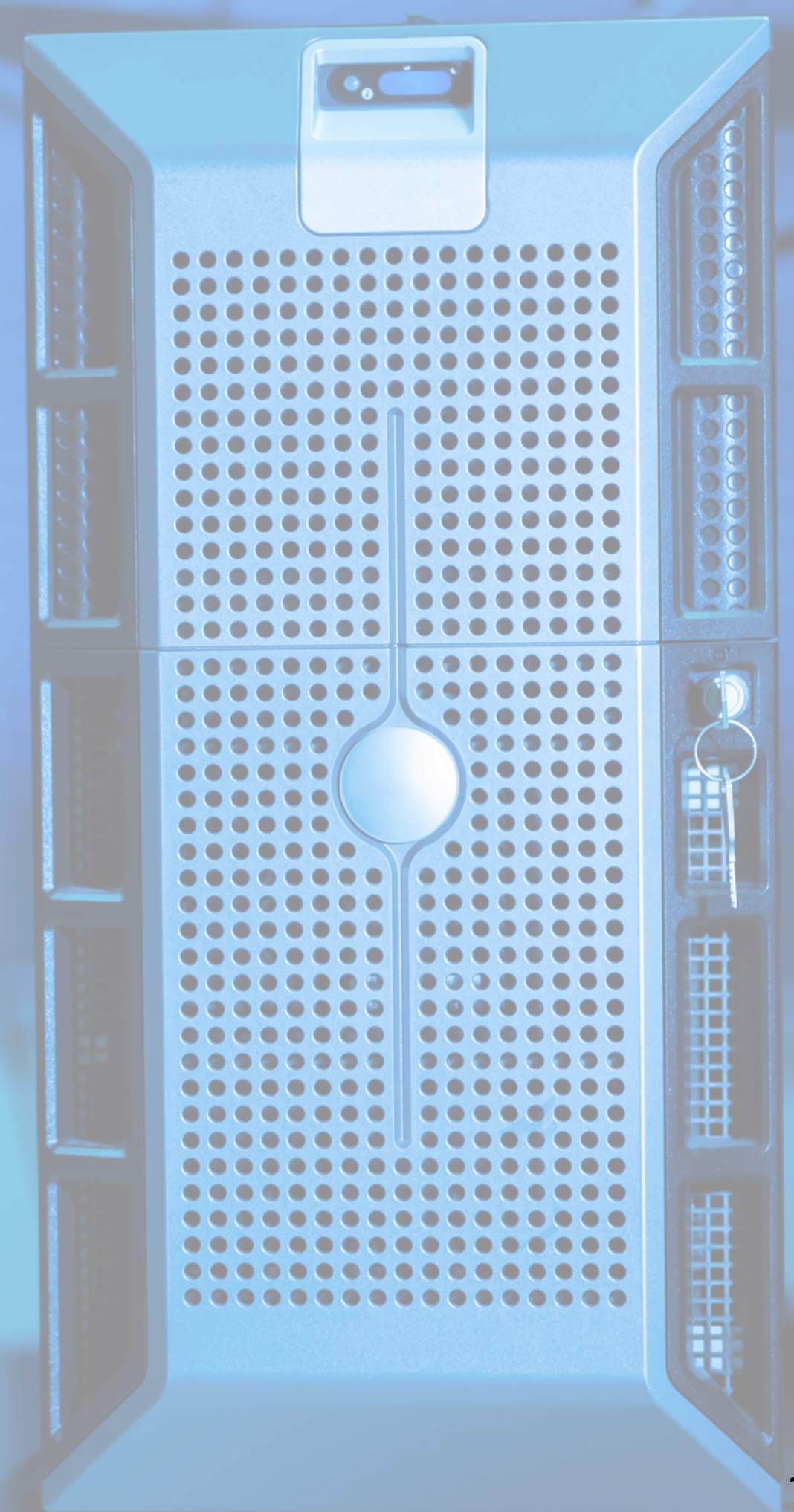
Phase 3: Commercial Rollout & National Scaling (Years 5+)

- Convert the successful pilot into the first commercial contract.
- Use this initial success as a powerful case study.
- Create a blueprint to systematically expand to power corporations across all of Vietnam.

Quantum Revolution

It's **not necessarily** the strongest or most intelligent that survives, but the **most adaptable to change**.

Charles Darwin



Backup Slide

QUANTUM POWER FLOW ALGORITHM

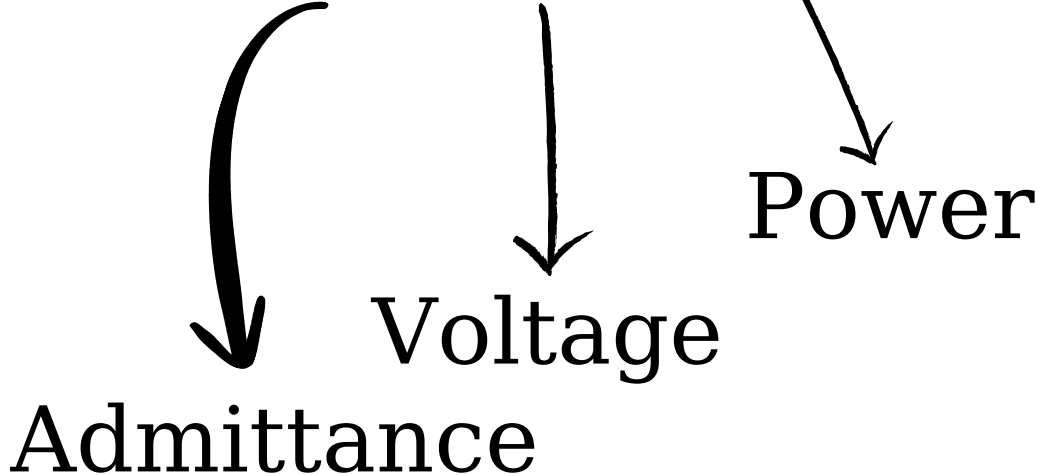
	3 bus	5 bus	9 bus	17 bus
B' matrix size	2x2	4x4	8x8	16x16
Circuit Width	5	7	9	11
Circuit Depth	336	3528	76876	802737
CNOT Gates	108	1181	28956	298594

Fast Decoupled Load Flow (FDLF)

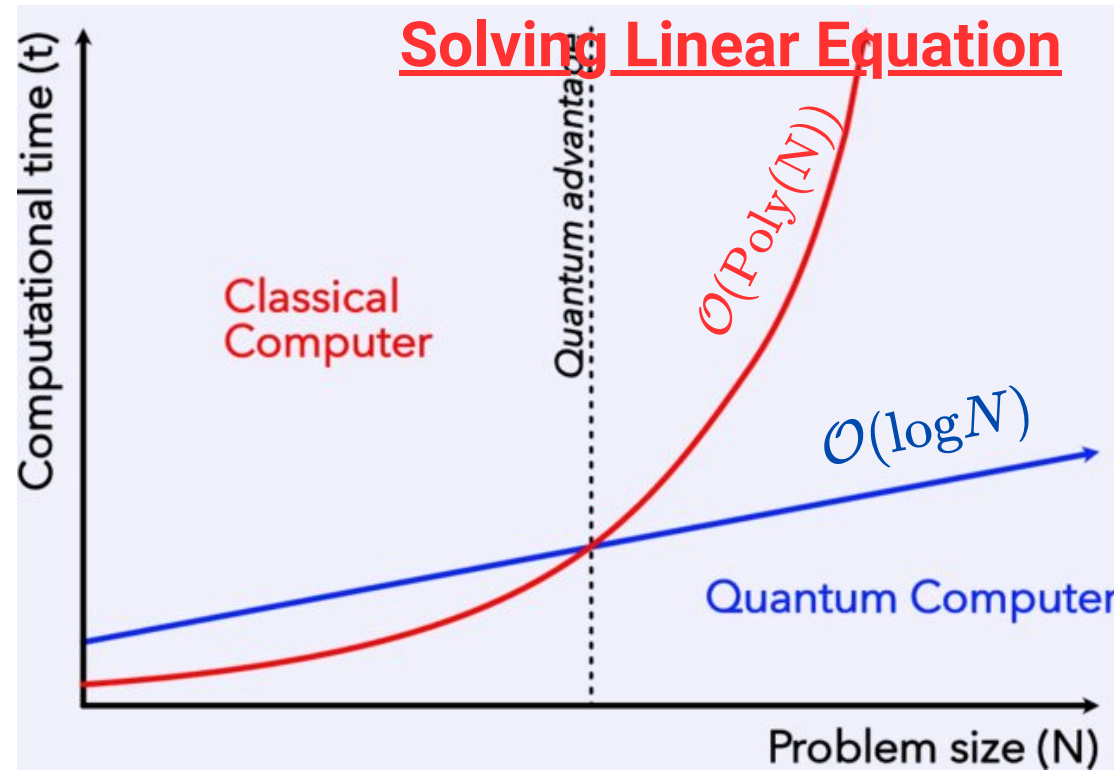
Solving Linear Equation

...

$$\mathbf{A} \cdot \mathbf{x} = \mathbf{b}$$

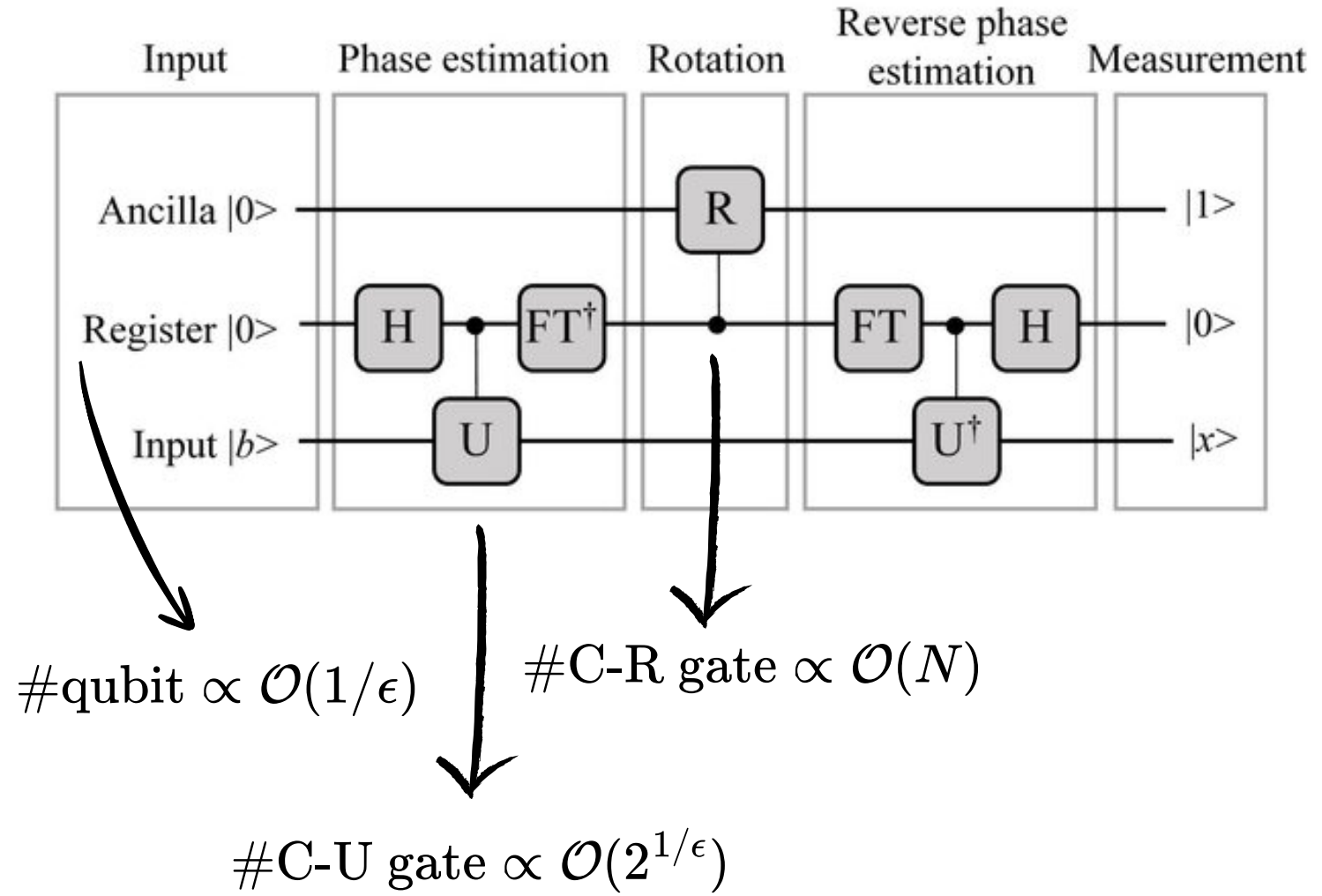


Quantum Power Flow (QPF)



Cre: [10.48550/arXiv.2504.03866](https://arxiv.org/abs/2504.03866)

Cre: [10.48550/arXiv.2010.02442](https://arxiv.org/abs/2010.02442)



QUANTUM POWER FLOW ALGORITHM

Fast Decoupled
Load Flow (FDLF)

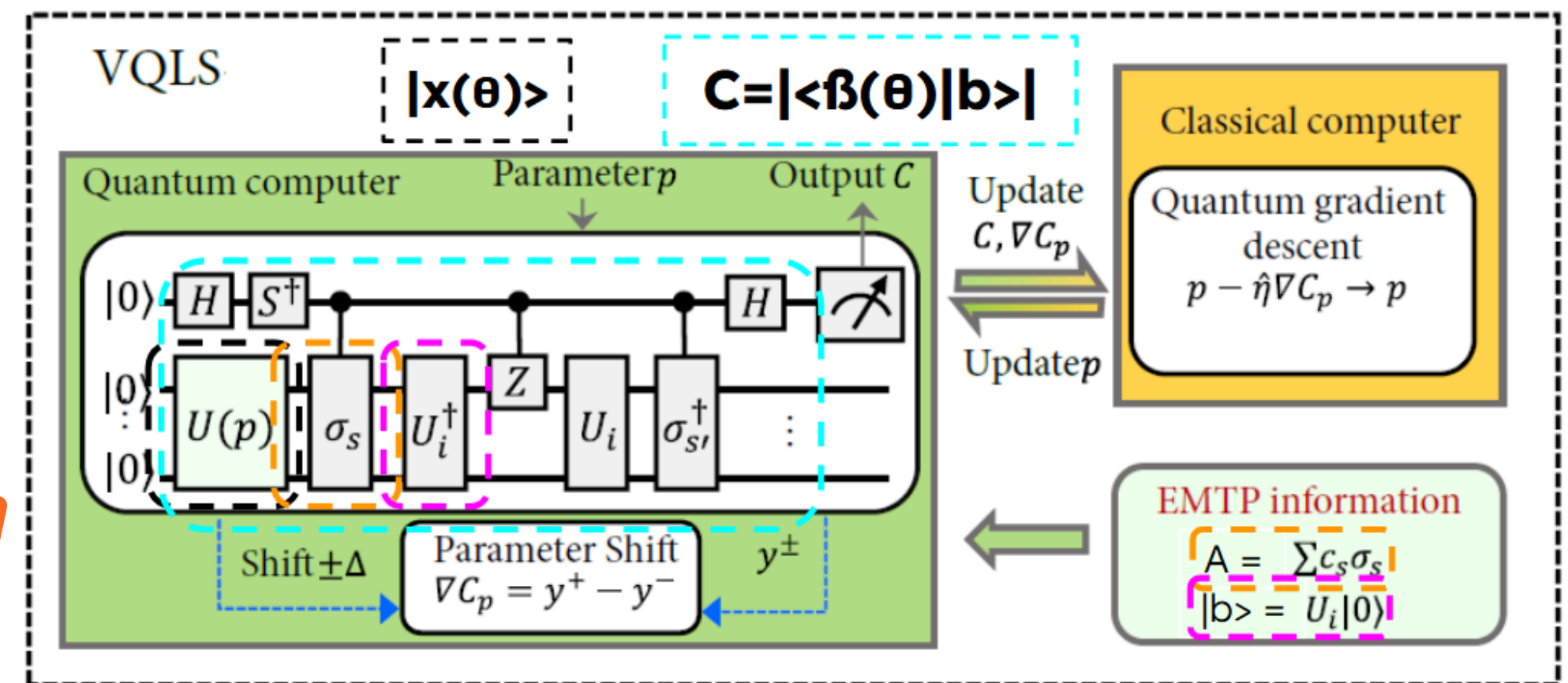
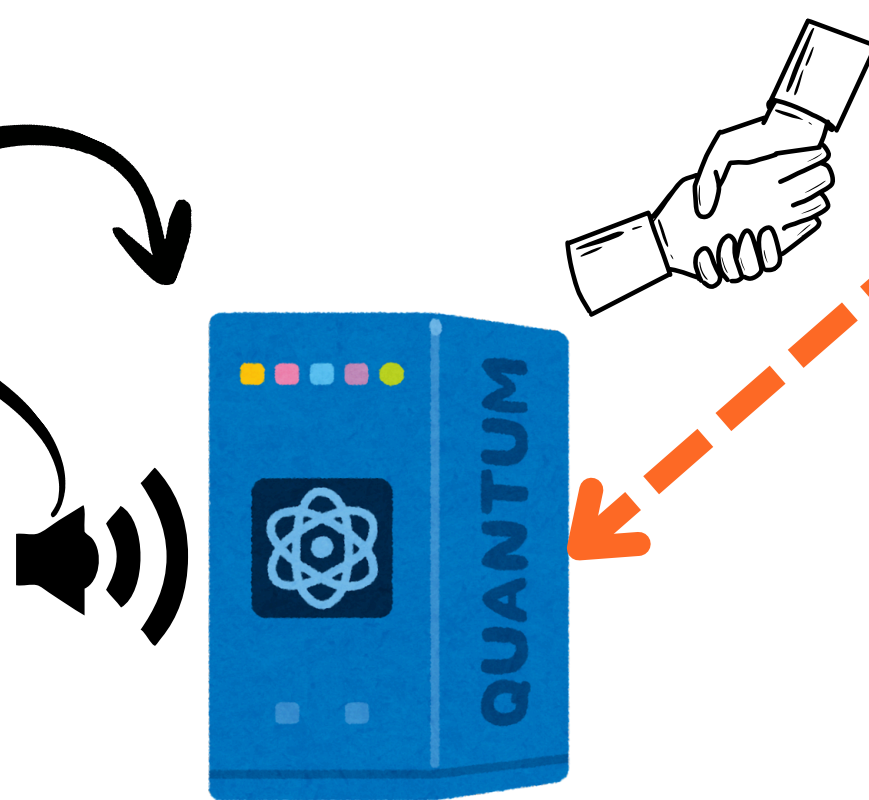
Variational Quantum Linear Solver (VQLS)

Solving Linear Equation

$$\mathbf{A} \cdot |\mathbf{x}(\theta)\rangle = |\mathbf{b}(\theta)\rangle$$

Stochastic
fluctuation

Renewable
Energy



Cre: 10.23919/IEN.2022.0021

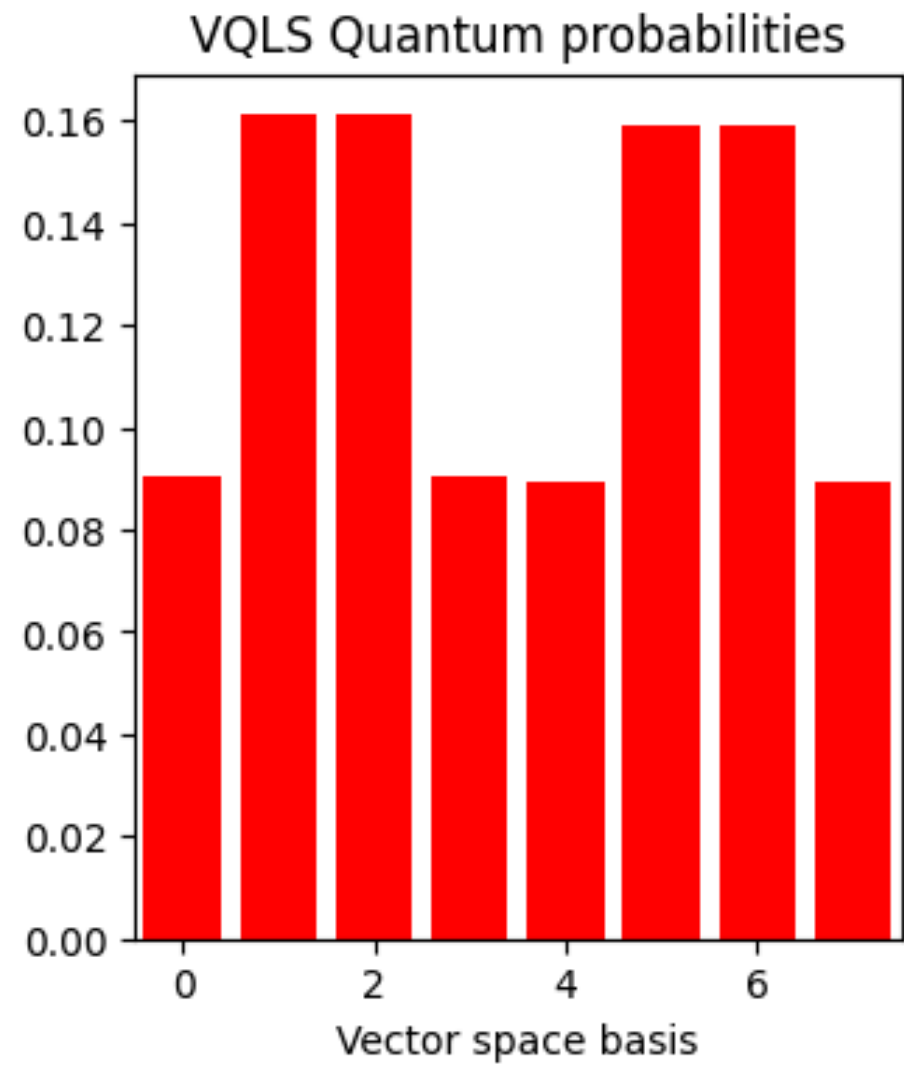
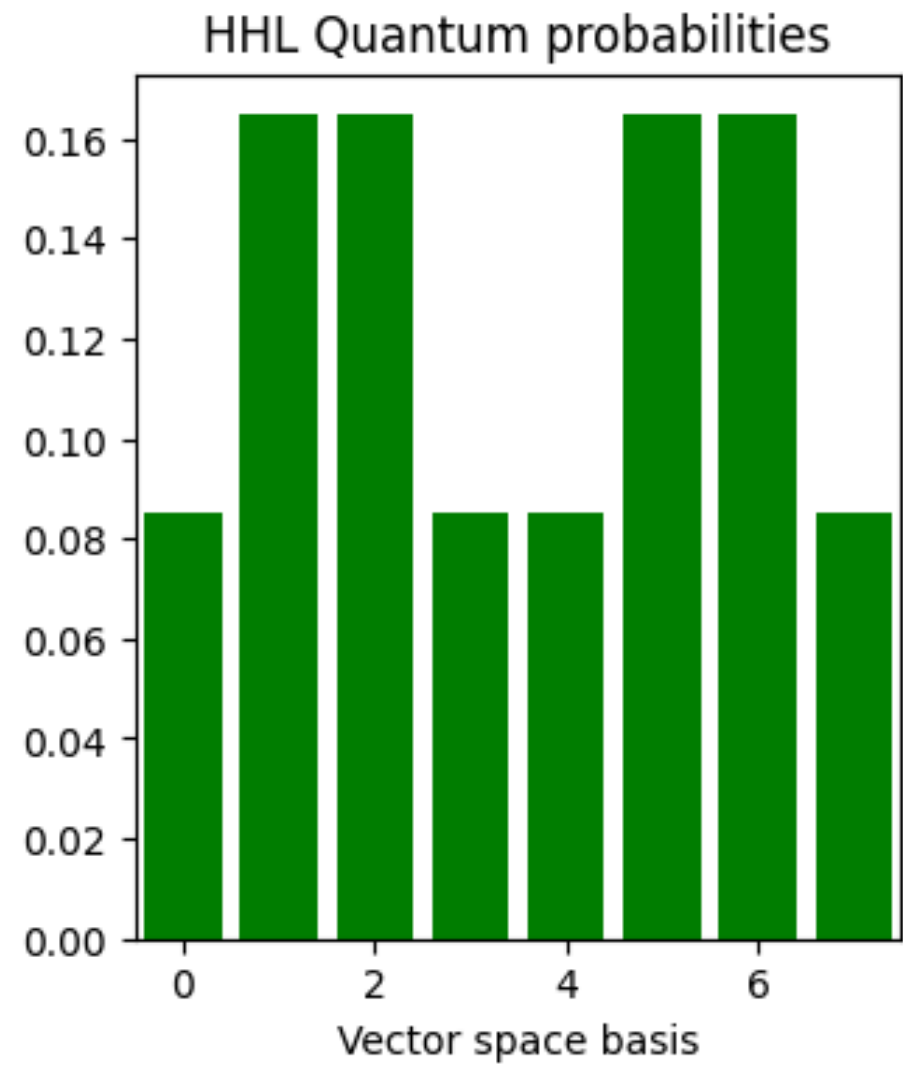
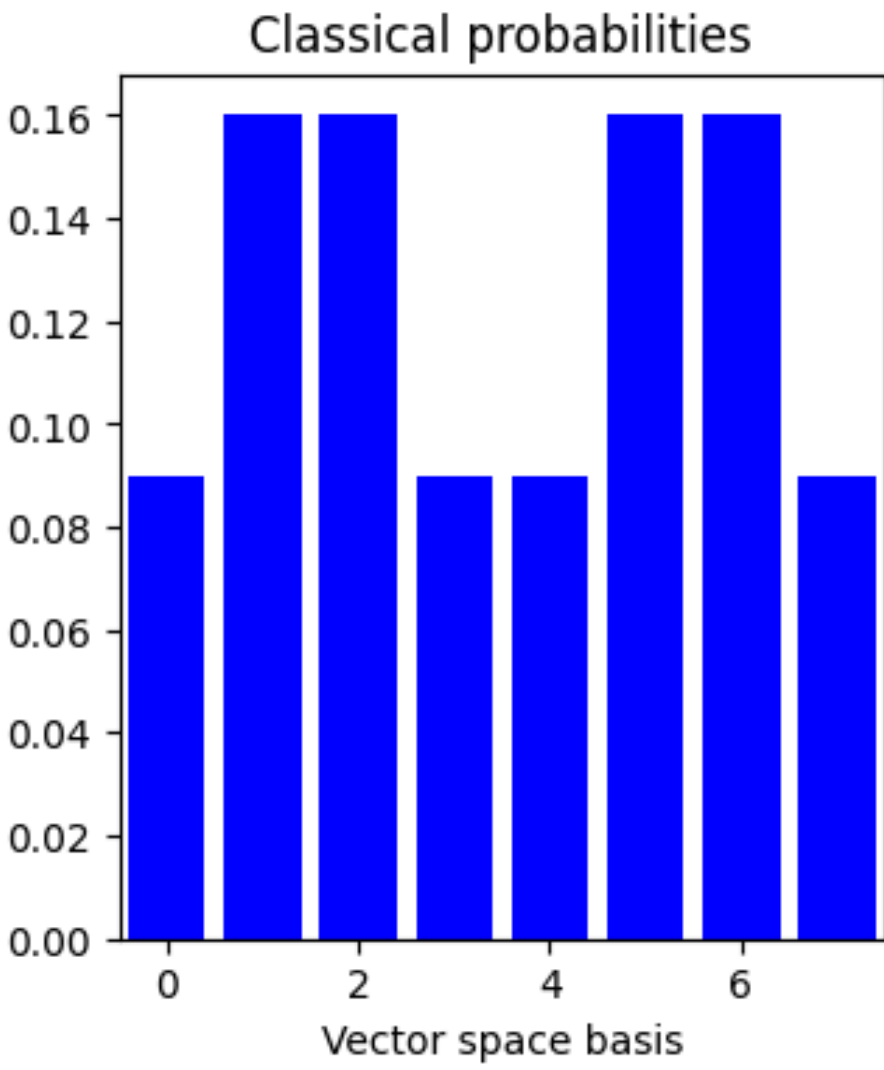
$$B'' = \begin{bmatrix} 4 & -1 & 0 & 0 & -1 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & 4 & -1 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 4 & 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 & 4 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & -1 & 4 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 & -1 & 4 & -1 \\ 0 & 0 & 0 & -1 & 0 & 0 & -1 & 4 \end{bmatrix}$$

Nine-buses system

RE: 0.00077

RE: 2.15e-5

VQLS



- Better than HHL (30 times)
- Reduce the quantum resources (~99.98%)

- CNOT: 54004
- Depth: 151795
- #qubit: 8

- 100 iterations
- CNOT: 4
- Depth: 24
- #qubit: 4